

Tejasvi Kothapalli

✉ tejasvi.kothapalli@berkeley.edu ☎ 408-802-0896

Education

Aug 2023 – Present	University of California, Berkeley – Vision Science Ph.D. Area: Computational Neuroscience & NeuroAI Advisor: Jacob Yates, Thesis Committee: Bruno Olshausen, Jack Gallant, Alex Huk, Jacob Yates Labs: The Active Vision and Neural Computation Lab (Led by Jacob Yates), The Redwood Center for Theoretical Neuroscience (Directed by Bruno Olshausen) Cumulative GPA: 3.979 Course Work: VS 260A: Optical and Neural Limits to Vision (A), VS 260C: Visual Neuroscience (A), VS 260D: Seeing in Time, Space and Color (A+), VS 265: Neural Computation (A)
Aug 2018 – May 2022	University of California, Berkeley – Electrical Engineering & Computer Science B.S. Cumulative GPA: 3.746 Upper Division Technical GPA: 3.909 Upper Division Coursework: Math 110: Linear Algebra (A-), EE 120: Signals and Systems (P), EE 126: Probability and Random Processes (P), EE 127: Optimization Models in Engineering (A-), CS 100: Principles & Techniques of Data Science (A), CS 161: Computer Security (A), CS 170: Efficient Algorithms and Intractable Problems (A), CS 182: Designing, Visualizing and Understanding Deep Neural Networks (A-), CS 188: Introduction to Artificial Intelligence (A-), CS 189: Introduction to Machine Learning (A), CS 194-26: Intro to Computer Vision and Computational Photography (A), CS 194-80: Full Stack Deep Learning (A), CS H196A: Senior Honors Thesis Research (A+), CS 280: Computer Vision (A), CS 285: Deep Reinforcement Learning (A), CS 288: Natural Language Processing (A) Research: Worked with Professor Stella Yu, Professor Meng Lin, Professor Yubei Chen, Postdoc Peter Wang. Senior Honors Thesis: Studying Dry Eye Syndrome with Machine Learning.

Experience

Aug 2023 - Present	Graduate Student Researcher in The Active Vision and Neural Computation Lab Studying neural recordings from free-viewing primates. Building a foundation model for Primate V1. Also modeling neural recordings with Sparse Coding.
Jun 2020 - May 2023	Researcher at ICSI (International Computer Science Institute) The Effects of Whitening in Neural Networks: We modified Batch Normalization to decorrelate the feature map. We found whitening to improve ResNet training on CIFAR-100 by over 2 percent. Collaborated with Professor Yu, Yubei Chen, and Peter Wang. Meibography Artificial Intelligence: Used computer vision and classical machine learning techniques to predict eye diseases. Collaborated with Professor Yu, Professor Lin, and Peter Wang.
Aug 2021 - May 2023	Student Research Assistant at CRC (Clinical Research Center) EasyTear Lipid Layer Analysis: Using computer vision techniques on videos of eye to determine lipid layer motion and thickness. Collaborated with Professor Lin, Professor Yu, Peter Wang.
Aug 2022 - Dec 2022	Machine Learning Engineer at Aizip Startup in the tinyML space where ML models are deployed to IoT devices. Worked on people detection and fall detection.
May 2017 - Aug 2017	NASA Ames Research Center Intern Worked in the Tensegrity Robot Division. Contributed to an open source web based tensegrity robot simulator. Used machine learning evolutionary algorithm to locomote twelve-rod tensegrity structures in simulation.

Publications

- 1 | **Artificial Intelligence Models Utilize Lifestyle Factors to Predict Dry Eye-Related Outcomes**
Andrew Graham, Jiayun Wang, **Tejasvi Kothapalli**, Jennifer Ding, Helen Tasho, Alisa Molina, Vivien Tse, Sarah M. Chang, Stella X. Yu, Meng C. Lin. *Nature Scientific Reports*, 2025
- 2 | **A Machine Learning Approach to Predicting Dry Eye-Related Signs, Symptoms and Diagnoses from Meibography Images**
Andrew Graham, **Tejasvi Kothapalli**, Jiayun Wang, Jennifer Ding, Vivien Tse, Penny A. Asbell, Stella X. Yu, Meng Lin. *Heliyon*, 2024
- 3 | **Tracking the Dynamics of the Tear Film Lipid Layer**
Tejasvi Kothapalli, Charlie Shou, Peter Wang, Tatyana Svitova, Andrew Graham, Meng Lin, Stella Yu. *Workshop at Neural Information Processing Systems (Neurips): Medical Imaging*, 2022
- 4 | **Saving Energy in Homes Using Wi-Fi Device Usage Patterns**
Tejasvi Kothapalli. *International Journal of Energy Optimization and Engineering (IJEEO)*, 2018
- 5 | **Controlling Tensegrity Robots through Evolution using Friction based Actuation**
Tejasvi Kothapalli, Adrian Agogino. *NASA Technical Reports*, 2017

Abstracts

- 1 | **Standard models fail to capture positional invariances in marmoset foveal V1**
Tejasvi Kothapalli*, Ryan Ressmeyer*, Declan Rowley, Jude Mitchell, Jacob Yates. *Society for Neuroscience*, 2025
- 2 | **Lifestyle and behaviors: predicting clinical signs and symptoms with machine learning**
Meng C. Lin, Andrew D. Graham, **Tejasvi Kothapalli**, Peter Wang, Jennifer Ding, Vivien Tse, Stella X. Yu. *Association for Research in Vision and Ophthalmology (ARVO)*, 2023
- 3 | **AI provides deeper understanding of Meibomian gland morphology and function**
Meng C. Lin, Peter Wang, **Tejasvi Kothapalli**, Andrew D. Graham, Stella X. Yu. *American Academy of Optometry (AAOPT)*, 2022

Honors, Fellowships & Selected Programs

- | | |
|---------------------|---|
| Aug 2026 | MIT Brains, Minds and Machines Summer Course
Selected to attend a fully funded intensive course at MBL Woods Hole focused on neuroscience, cognitive science, computer science, and biologically inspired AI. |
| July 2026 | CSHL Computational Neuroscience: Vision Course
Selected to attend a long-running Cold Spring Harbor Laboratory course in computational visual neuroscience, focused on neural coding, decoding, attention, decision-making, and perception. |
| Aug 2023 - May 2025 | CIVO Fellowship
Generously funded for graduate studies to conduct research which promotes innovative display, graphics, and optical technology for the healthy and diseased eye. |
| Aug 2021 - May 2022 | EECS Honors Program
Recognizes EECS students who commit to research, strong academics, and writing a senior thesis. |

Teaching & Services

Aug 2025 - Dec 2025	GSI for Neuroscience 172L: Cognitive and Computational Lab Course teaches experimental and analytical techniques used by cognitive and computational neuroscientists. Led lab sections, content creation, grading, and office hours.
Aug 2023 - September 2024	Program Committee Chair, Bay Area Vision Research Day (BAVRD) Help plan, fundraise, and invite speakers and abstract presenters.
Aug 2023 - Present	CIVO BASIS Coordinator I help organize and teach about general Vision Science to 4th graders around the East Bay. We work with the organization Bay Area Scientists Inspiring Students (BASIS).
May 2022 - Present	Teacher at Inspirit AI Teaching machine learning to high students at Khan Lab School, Bellarmine College Preparatory, Evergreen Valley High School, and Bentley School.

Talks

December 2025	CIVO Day: Measuring Sensitivity of V1 activity to eye movements
October 2025	Vision Science Retreat: Standard models fail to capture positional invariances in marmoset foveal V1
September 2025	Bay Area Vision Research Day (BAVRD): Towards a Foundation Model for Primate Visual Cortex
May 2025	Redwood Meeting: V1 Sensitivity to Eye Movements and Predicting V1 with Sparse Coding
December 2024	CIVO Day: High-accuracy retinal input tracking in free-viewing primates

References

- 1 | **Jacob Yates**, yates@berkeley.edu
Professor, Herbert Wertheim School of Vision Science and Optometry, UC Berkeley
- 2 | **Bruno Olshausen**, baolshausen@berkeley.edu
Professor, Neuroscience and Herbert Wertheim School of Optometry & Vision Science, UC Berkeley
Director, Redwood Center for Theoretical Neuroscience
- 3 | **Alex Huk**, alexhuk@g.ucla.edu
Professor, Departments of Psychiatry and Ophthalmology, David Geffen School of Medicine, UCLA
Director, Fuster Laboratory
- 4 | **Peter Wang**, peterwg@berkeley.edu
Postdoctoral Scholar, California Institute of Technology
- 5 | **Michael Silver**, masilver@berkeley.edu
Professor, Neuroscience and Herbert Wertheim School of Optometry & Vision Science, UC Berkeley
Director, UC Berkeley Center for the Science of Psychedelics
- 6 | **Frederic Theunissen**, theunissen@berkeley.edu
Professor, Neuroscience and Integrative Biology, UC Berkeley